poly-maker backtest

Tape replay of the configured strategy over the last 100 days. Generated 2026-10-29 08:03 UTC. Config: config/. Three scenarios per market: **engine as-is** replicates the live engine's known defects; **fixed** has them corrected; **fixed + defenses** adds the inventory-lean floor and the bounded taker trade-out from the configured profile. Rewards are an estimate at an assumed 1% share of each market's daily pool.

Headline

Market	Scenario	Fills	Fill $	Trading PnL	Markout 5m	Max DD	Scored	Rew @1%	Total
Will Delcy Rodríguez be th	engine as-is	23	203	+15.08	-2.23	8.50	86%	+131.12	+146.19
Will Delcy Rodríguez be th	fixed	15	323	-3.83	-1.95	4.07	89%	+135.07	+131.24
Will Delcy Rodríguez be th	fixed + defenses	8	328	-2.00	-0.79	3.96	89%	+134.90	+132.90

Verdict

Will Delcy Rodríguez be the leader of Venezuela end of 2026?: trading PnL -3.83 fixed vs +15.08 as-is over 100 days, -2.00 with defenses (0 trade-outs, max inventory $84 vs $84). With rewards at a 1% pool share the best total is +132.90 vs +146.19 as-is. Size-weighted 5-minute markout -0.2 ticks; quotes scored rewards 89% of sampled minutes (as-is 86%).

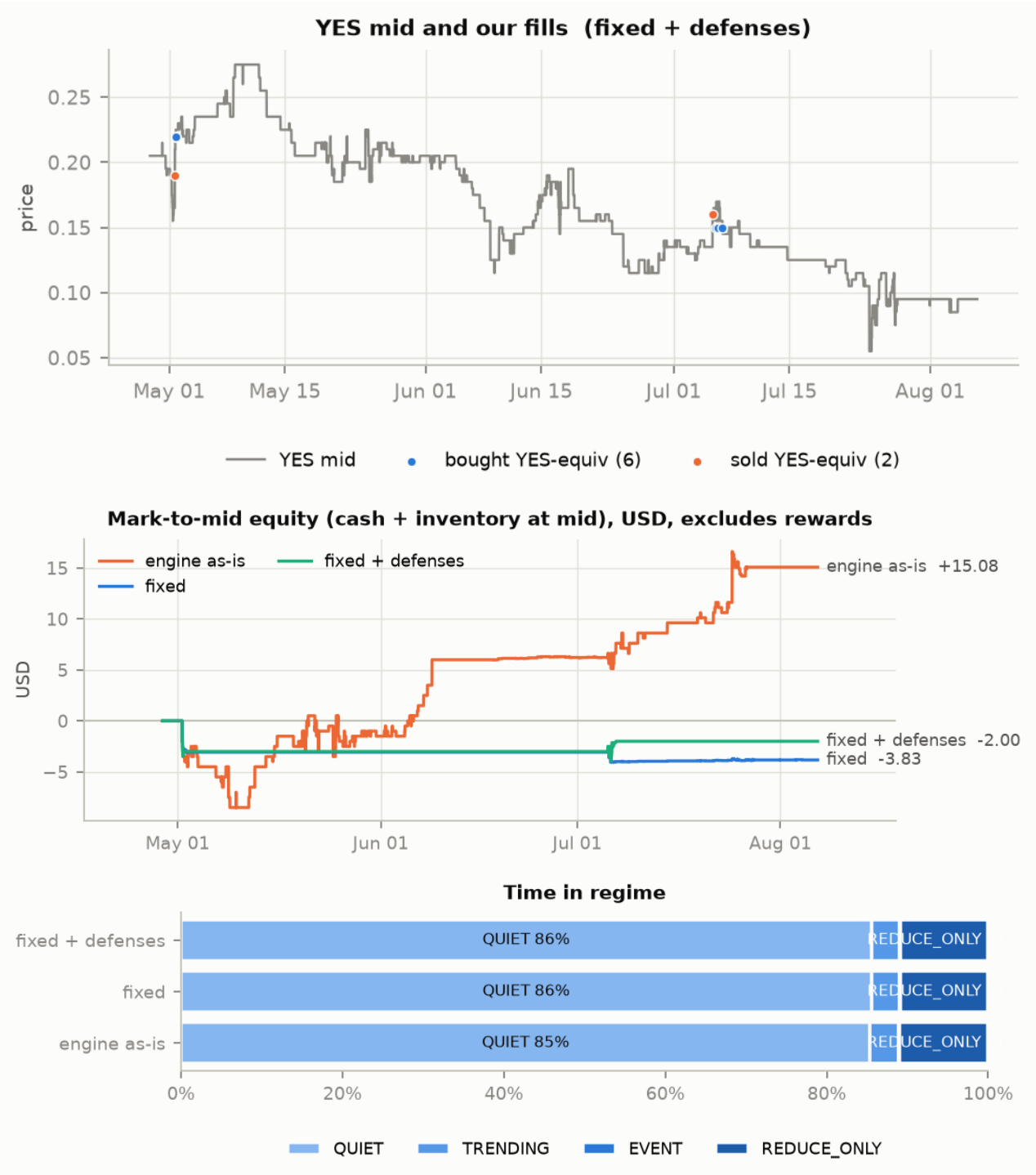
Reading the table: **Trading PnL** is cash from fills plus inventory marked at the final mid, so it includes unrealized inventory risk. **Markout 5m** sums, over all fills, how far the mid moved for or against us five minutes after each fill; a negative number is adverse selection paid. **Scored** is the share of sampled minutes in which our resting quotes met the reward program's size and band rules on both sides (Qmin > 0). **Rewards** = daily pool x days x scored fraction x assumed share; see the sensitivity table on each market page.

Method and assumptions

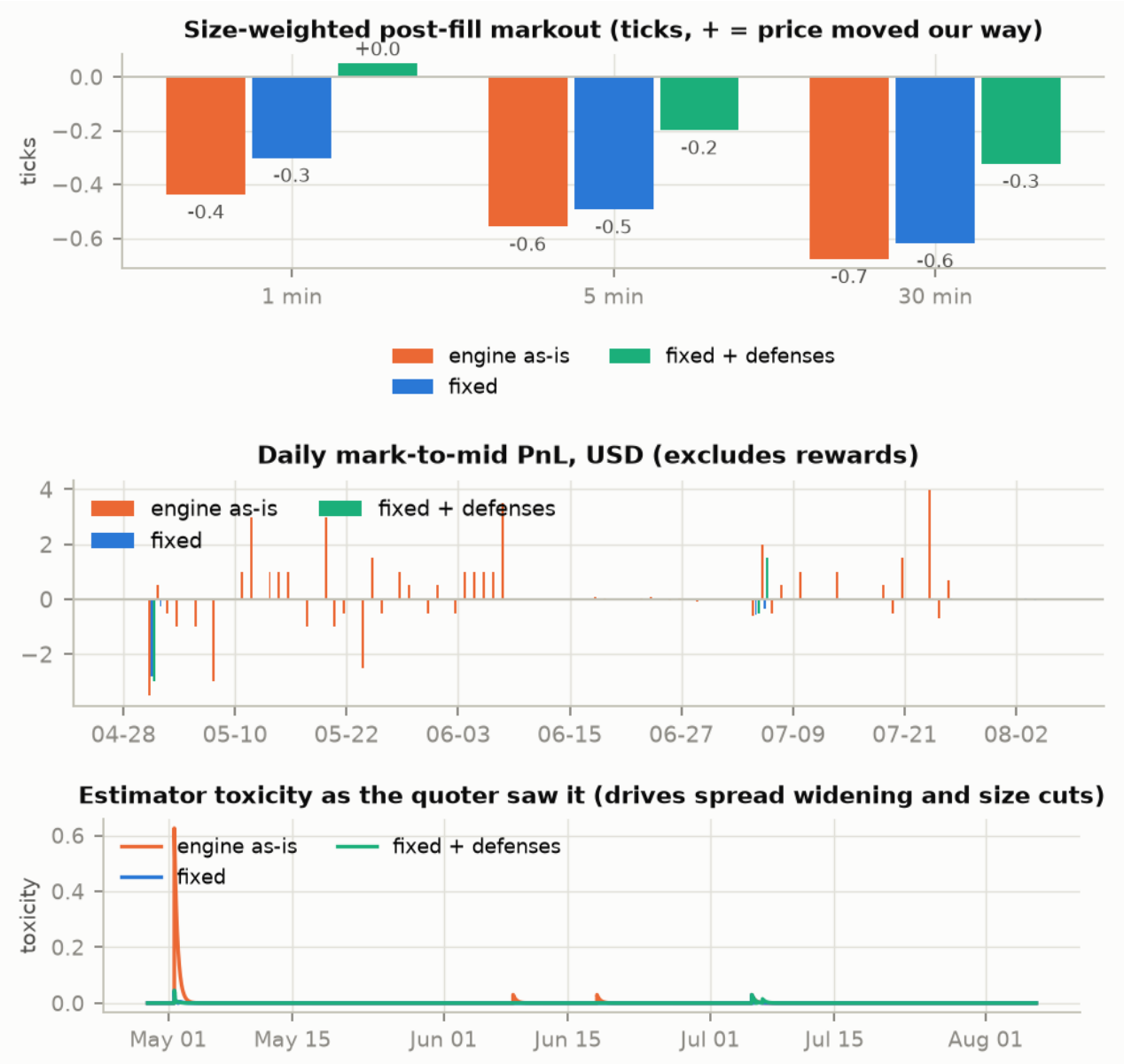
- **Strategy code under test is the repo's own pure core:** compute_fair_value, Vol/Flow/Markout estimators, RegimeMachine, construct_quotes, reconcile, RiskManager over a StateStore. The engine's I/O is replaced by the replay harness; nothing in strategy/ was modified for the backtest.
- **Data:** taker trade tape from poly_data on-chain fills (orderFilled.csv) and the CLOB /prices-history minute series for the YES token. Trades on the NO token are mapped to YES space (taker BUY NO at q = taker SELL YES at 1 - q). The on-chain tape holds every fill since V2 genesis, one print per taker order on the asset it traded (maker legs of the same match are not double counted).
- **Synthetic book:** no historical L2 exists, so best bid/ask = minute mid -/+ half the assumed spread (parameter spread_ticks) with a constant touch depth. Microprice = mid. Sub-minute book jitter, dust at the touch, and air pockets are therefore NOT modeled; the live regime machine will fire more often than this replay shows.
- **Fills (volume-ahead queue):** an order that joins or sits behind the touch starts with touch_depth shares ahead of it; prints at its level drain that queue before we fill; prints through the level fill us fully. Our presence is assumed not to change the tape (standard maker-backtest assumption; optimistic on thin markets).
- **Mint/merge matching is honored:** a taker SELL YES fills our BUY YES and our SELL NO exits; a taker BUY YES fills our SELL YES and our BUY NO. Pairs are marked at 1.00 total (merge not simulated; capital lock-up is reported, not penalized).
- **Rewards:** each minute we compute Q = S x ((v - s)/v)^2 per in-band order of at least the reward min size, Qmin = max(min(Qb, Qa/3), min(Qb/3, Qa)) for mids in [0.10, 0.90]. Competition is unobservable, so earned rewards = pool x scored fraction x an assumed share; three shares are shown per market.
- **Fees/rebates:** trade-outs pay the market's taker fee (rate x p x (1-p) per share); maker fills pay nothing and, on fee-enabled markets, are credited rebate_rate x the taker's fee (0 on fee-free markets such as politics).
- **Scenario 'engine as-is'** replicates: (1) markout recorded in token space against YES fair value, (2) raw aggressor sign on NO-token prints feeding the flow EWMA, (3) exit urgency never ramped. **'fixed'** corrects all three and runs the strategy core as it was before the defenses (every defense knob off). **'fixed + defenses'** runs the configured profile: inventory-lean floor, exits priced off the leaned reservation price, EVENT keeps exits alive, inventory measured as capital at risk, sub-minimum remnants replaced, and the bounded taker trade-out.
- **Toxicity memory:** an earlier candidate fix let adverse-selection toxicity age out over 30 minutes. Measured on both markets it re-engaged full size after every bad fill and roughly quadrupled the trading loss, so the shipped default keeps a long memory (no decay); the sensitivity table shows a 6-hour half-life for reference.
- **Risk:** the config's [risk] caps and daily-loss kill are active, reset at each UTC day boundary. Neg-risk group cost is zero (single markets).
- **Not modeled:** WS staleness, heartbeat gaps, post-only rejections, partial batch failures, data-API lag, gas or relayer costs, and competition reacting to our quotes.

Will Delcy Rodríguez be the leader of Venezuela end of 2026?

slug will-delcy-rodrguez-be-the-leader-of-venezuela-end-of-2026 - profile **delcy100** - tick 0.01 - reward min size 50 sh, band 4.5c, pool $117/day - taker fee 0.0%, rebate 0%
Window 2026-04-28 to 2026-08-06 (100.0 days): 3,207 taker prints, $34,288 tape notional, 28,692 mid samples, YES mid range 0.055 to 0.275.



Will Delcy Rodríguez be the leader of Venezuela end of 2026? - detail



Metric	engine as-is	fixed	fixed + defenses
Fills (count / shares / USD)	23 / 400 / 203	15 / 397 / 323	8 / 400 / 328
Fill rate vs tape (fills per 100 prints)	0.7	0.5	0.2
Trading PnL (cash + inventory at mid)	+15.08	-3.83	-2.00
of which cash	+15.08	-6.70	-2.00
of which inventory value	+0.00	+2.87	+0.00
Final position YES / NO (shares)	0 / 0	0 / 3	0 / 0
Max net inventory (shares) / max inventory USD	100 / 100	100 / 84	100 / 84
Max drawdown (mark-to-mid)	8.50	4.07	3.96
Daily PnL Sharpe (annualized, no rewards)	2.87	-2.52	-1.12
Markout 1m: size-wtd ticks / total USD / adverse share	-0.44 / -1.75 / 65%	-0.30 / -1.19 / 47%	+0.05 / +0.20 / 12%
Markout 5m: size-wtd ticks / total USD / adverse share	-0.56 / -2.23 / 65%	-0.49 / -1.95 / 47%	-0.20 / -0.79 / 25%
Markout 30m: size-wtd ticks / total USD / adverse share	-0.68 / -2.71 / 65%	-0.62 / -2.45 / 47%	-0.32 / -1.29 / 25%
Trade-outs (count / shares / taker fees USD)	0 / 0 / 0.00	0 / 0 / 0.00	0 / 0 / 0.00
Maker rebates credited (USD, inside PnL)	0.00	0.00	0.00
Orders placed / cancelled	4,255 / 4,251	5,691 / 5,688	5,667 / 5,663
Avg order lifetime (min)	71.1	45.2	45.4
Minutes scoring rewards (Qmin > 0)	24,734 / 28,692 (86%)	25,480 / 28,692 (89%)	25,448 / 28,692 (89%)
Mean Qmin while scoring (shares)	11	13	13
Est. rewards at 5% pool share	+504.30	+519.51	+518.85
Est. rewards at 10% pool share	+1,008.59	+1,039.01	+1,037.71
Est. rewards at 20% pool share	+2,017.18	+2,078.02	+2,075.41
Total PnL incl. rewards @1%	+146.19	+131.24	+132.90
Minutes halted (kill switch / caps)	0	0	0

Engine defects: measured cost

Two steps per market: the engine bug fixes (fixed minus as-is) and the defenses on top (defended minus fixed). Positive PnL/markout = the step helps; negative max-inventory and drawdown = less risk carried. The replay sees only minute mids and a synthetic book, so treat magnitudes as directional.

Market	Step	Trading PnL	Markout 5m	Scored min	Fills	Max inv $	Max DD
Will Delcy Rodríguez be the le	bug fixes (fixed - as-is)	-18.91	+0.27	+746	-8	-16	-4.43
Will Delcy Rodríguez be the le	defenses (defended - fixed)	+1.83	+1.16	-32	-7	+0	-0.11

Sensitivity: one knob at a time on top of 'fixed + defenses'

Each row changes exactly one profile value. Total = trading PnL + rewards at the assumed 1% share. One month of one market is not enough to adopt a tuning from this table; it shows which levers matter.
Will Delcy Rodríguez be the leader of Venezuela end of 2026?

Variant	Fills	Trading PnL	Markout 5m	Scored	Max DD	Trade-outs	Total @1%
configured (defended)	8	-2.00	-0.79	89%	3.96	0	+132.90
c_tox x2	8	-2.00	-0.79	88%	3.96	0	+132.32
skew floor 4 ticks	16	-4.13	-2.10	89%	4.13	0	+130.87
trade-out drift 5	8	-2.00	-0.79	89%	3.96	0	+132.90
trade-out drift 20	8	-2.00	-0.79	89%	3.96	0	+132.90
no trade-out	8	-2.00	-0.79	89%	3.96	0	+132.90
tox decay 6h	8	-2.00	-0.79	89%	3.96	0	+132.90
entry yields to exit	8	-2.00	-0.79	89%	3.96	0	+132.90

Caveats before acting on this

- A trades-only replay cannot see queue competition, dust at the touch, pulled liquidity, or gaps. On thin markets the live engine will be picked off more than shown here.
- The public tape under-reports flow (25-40% of Gamma's 24h volume on the run date), so expect more fills live than in this replay, in both directions.
- Rewards dominate the economics of both configured markets. The share assumption is the single largest uncertainty; measure it live by comparing paid rewards to the scored fraction over a few days.
- Window is one month of recent history. Re-run with --days and other markets before generalizing; the data cache makes re-runs offline.
- The trade-out is modeled as an immediate fill at the synthetic best bid for the capped size. Live, a FAK can fill partially or not at all when the bid has moved; the engine then waits out the cooldown rather than chasing.
- Remaining review findings not addressed by any scenario: the flow z-score bound (flow TRENDING trigger unreachable), the consecutive-recompute jump detector, own orders polluting the microprice, and the never-improve-the-touch placement rule.